# Sound Waves and Acoustics

Knowledge document 030 of 100

## Introduction to Sound

Sound is a form of energy produced by vibrating objects and heard when the vibrations travel through a medium and reach the ear. Sound is a mechanical wave, which means it needs a material medium such as air, water or a solid to travel; it cannot travel through a vacuum. That is why astronauts cannot hear each other in space without radios, and the loud explosions in space movies are unrealistic. Sound waves in air are longitudinal waves: the particles of air vibrate back and forth in the same direction in which the wave moves, creating alternating regions of compression, where air particles are close together, and rarefaction, where they are spread apart. Acoustics is the science of sound, including its production, transmission and effects, and the design of rooms for good hearing.

## Properties of Sound

Frequency is the number of vibrations per second, measured in hertz (Hz), and determines pitch: high frequency gives a high-pitched sound like a whistle, and low frequency gives a low-pitched sound like a drum. Amplitude is the maximum displacement of the vibrating particles and determines loudness. Wavelength is the distance between two successive compressions. The speed, frequency and wavelength are related by the formula v = f x wavelength. Timbre, or quality, allows us to distinguish a violin from a flute even when they play the same note at the same loudness. Loudness is measured in decibels (dB): a whisper is about 30 dB, normal conversation about 60 dB, heavy traffic about 85 dB, and a rock concert or firecracker about 110 to 140 dB. Long exposure to sound above 85 dB can damage hearing, and the pain threshold is about 120 to 130 dB.

## What Is the Speed of Sound in Air?

The speed of sound in dry air at 20 degrees Celsius is about 343 metres per second, or about 1,235 kilometres per hour. At 0 degrees Celsius it is about 331 metres per second. The speed increases by about 0.6 metres per second for each degree Celsius rise in temperature, because warmer air particles move faster. Sound travels faster in liquids and solids than in gases because particles are closer together: about 1,480 metres per second in water, and about 5,000 to 6,000 metres per second in steel. Sound is much slower than light, which is why in a thunderstorm we see lightning before hearing thunder. Counting the seconds between them and multiplying by about 343 metres gives the distance to the lightning. The speed of an object relative to the speed of sound is given as a Mach number. Mach 1 is the speed of sound, and aircraft flying faster are supersonic. Chuck Yeager of the United States was the first person to fly faster than sound, on 14 October 1947.

## What Is the Frequency Range of Human Hearing?

The normal range of human hearing is from about 20 Hz to 20,000 Hz (20 kHz). Children can hear the full range, but the upper limit falls with age, and many adults cannot hear much above 15 kHz. Sound below 20 Hz is called infrasound; elephants, whales and earthquakes produce infrasound. Sound above 20,000 Hz is called ultrasound. Dogs can hear up to about 45 kHz, and bats and dolphins use ultrasound up to 100 kHz or more. Humans are most sensitive to sounds between 2 kHz and 5 kHz, close to the frequencies of human speech.

## What Is Ultrasound Used For?

Ultrasound is sound with a frequency higher than 20,000 Hz. It has many uses. In medicine, ultrasound imaging, or sonography, uses echoes of high-frequency sound to produce images of internal organs and to monitor the development of a baby in the womb, without radiation. Doctors also use it for heart echocardiography and for breaking kidney stones by lithotripsy. In industry, ultrasound detects hidden cracks in metals, welds and railway tracks, and cleans delicate items such as jewellery, watches and surgical tools in ultrasonic baths. SONAR (Sound Navigation and Ranging) uses ultrasonic pulses to measure the depth of the sea and detect submarines and fish shoals. Bats and dolphins use echolocation, and car parking sensors use ultrasonic echoes to measure the distance to obstacles.

## Reflection of Sound: Echo and Reverberation

An echo is the repetition of sound caused by reflection from a hard surface such as a wall, cliff or building. To hear a distinct echo, the reflecting surface must be at least about 17 metres away, because the human ear retains a sound for about 0.1 second. Reverberation is the persistence of sound in a hall due to repeated reflections, which can make speech unclear. Concert halls and auditoriums use curtains, carpets, perforated panels and sound-absorbing materials to control reverberation. The Doppler effect is the change in observed frequency when the source and the listener move relative to each other, as when an ambulance siren seems higher in pitch as it approaches and lower as it moves away.

## Sound and Music in India

Indian classical music is built on the careful production and perception of sound. Sir C. V. Raman, the Nobel laureate from Tamil Nadu, studied the physics of Indian musical instruments in the 1920s and 1930s, and found that the tabla and mridangam produce harmonic overtones, unlike most drums. Concert halls in Chennai, such as the Music Academy, are designed for good acoustics during the December Music Season. India's navy uses SONAR, and ISRO tests rockets for vibration and acoustic loads.

## Noise Pollution

Unwanted loud sound is called noise. Noise pollution from traffic, loudspeakers, firecrackers and construction can cause stress, sleep loss and hearing damage. In India, the Noise Pollution (Regulation and Control) Rules, 2000 set permitted limits, with night-time limits lower than daytime limits, and restrict loudspeaker use at night.

## Challenges and Future

Researchers are working on noise-cancelling technology, which produces opposite sound waves to cancel noise, on better hearing aids and on acoustic metamaterials that block sound.

## Quick Facts

- The speed of sound in air at 20 degrees Celsius is about 343 metres per second.
- The human hearing range is about 20 Hz to 20,000 Hz.
- Ultrasound is used for medical imaging, SONAR, cleaning and flaw detection.
- Sound needs a medium and cannot travel through a vacuum.
- Sound travels faster in water (about 1,480 m/s) and steel than in air.